Experiment-6

Write A Python Program For Vacuum Cleaner Problem

Aim

To write and execute a Python program to solve the Vacuum Cleaner Problem using a rule-based approach to clean all dirty rooms.

Algorithm

1. Start the program.
2. Initialize two rooms, A and B, as dirty.
3. Place the vacuum cleaner in Room A.
4. Check the current room's condition.
5. If the room is dirty, clean it and update its status to clean.
6. Move the vacuum cleaner to the other room.
7. Repeat the cleaning process until both rooms are clean.
8. Display the final status of both rooms.
9. Stop the program.

Code

Ex-6 AI.py - C:/Users/Dinesh/AppData/Local/Programs/Python/Python310/Ex-6 AI.py (3.10.0)

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```
# Vacuum Cleaner Problem

def vacuum_cleaner():
    room = {
        'A': 'Dirty',
        'B': 'Dirty'
    }

    position = 'A'

    print("VACUUM CLEANER PROBLEM")
    print("----------------------")

    while 'Dirty' in room.values():
        print("\nVacuum is in Room", position)

        if room[position] == 'Dirty':
            print("Room", position, "is Dirty")
            room[position] = 'Clean'
            print("Cleaning Room", position)

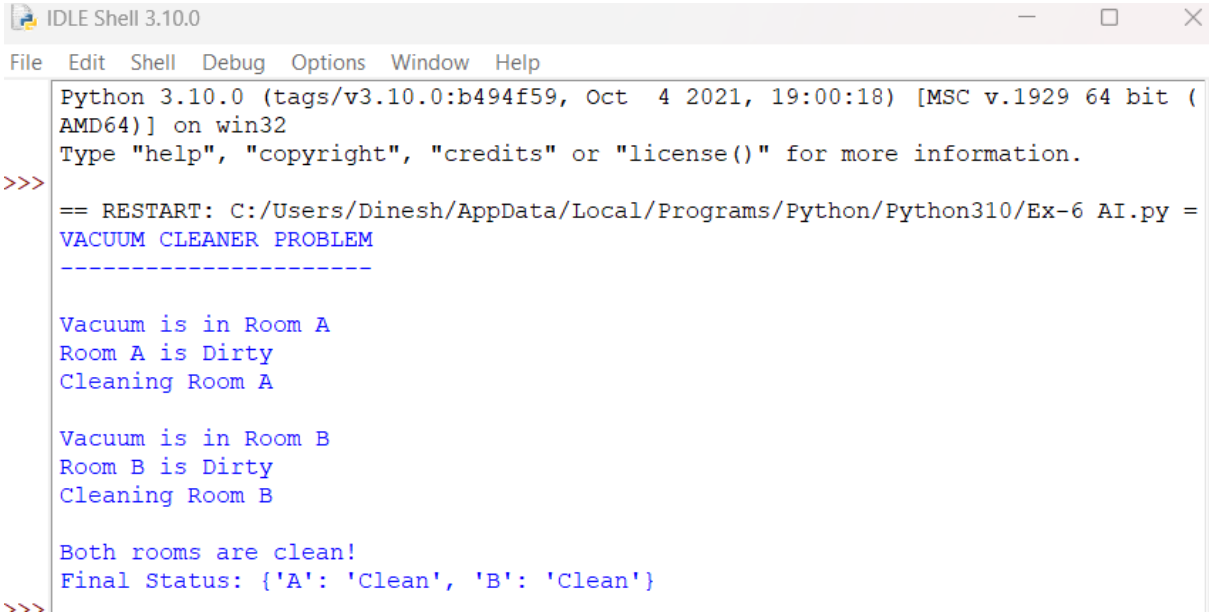
        else:
            print("Room", position, "is already Clean")

        if position == 'A':
            position = 'B'
        else:
            position = 'A'

    print("\nBoth rooms are clean!")
    print("Final Status:", room)


vacuum_cleaner()
```

Output



Result

The Vacuum Cleaner Problem was successfully implemented using Python 3.10. The program cleans both Room A and Room B and displays their final status as clean.